

Improving Low-Precision Network Quantization via Bin Regularization

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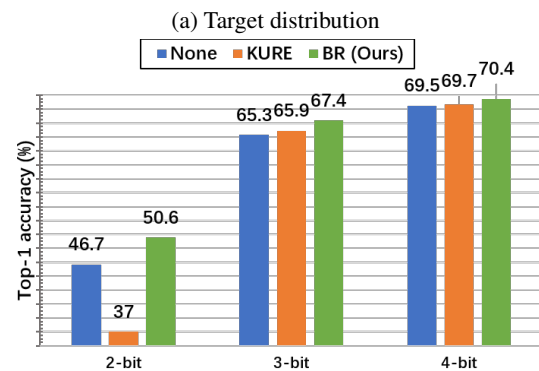
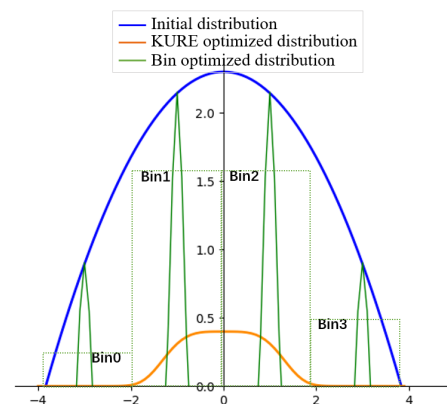
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Abstract

Model quantization is an important mechanism for energy-efficient deployment of deep neural networks on resource-constrained devices by reducing the bit precision of weights and activations. However, it remains challenging to maintain high accuracy as bit precision decreases, especially for low-precision networks (e.g., 2-bit MobileNetV2). Existing methods have been explored to address this problem by minimizing the quantization error or mimicking the data distribution of full-precision networks. In this work, we propose a novel weight regularization algorithm for improving low-precision network quantization. Instead of constraining the overall data distribution, we separately optimize all elements in each quantization bin to be as close to the target quantized value as possible. Such bin regularization (BR) mechanism encourages the weight distribution of each quantization bin to be sharp and approximate to a Dirac delta distribution ideally. Experiments demonstrate that our method achieves consistent improvements over the state-of-the-art quantization-aware training methods for different low-precision networks. Particularly, our bin regularization improves LSQ for 2-bit MobileNetV2 and MobileNetV3-Small by 3.9% and 4.9% top-1 accuracy on ImageNet, respectively.

1. Introduction

Deep Convolutional Neural Networks (CNNs) have achieved remarkable progress in a wide range of applications including computer vision, natural language processing, speech recognition, etc. With the popularity of CNNs, there is an increasing demand for techniques to run networks efficiently on resource-constrained devices (e.g., mobile phones or FPGA). These techniques include network quantization, pruning, manual design of efficient architecture, or neural architecture search (NAS). We focus on network quantization by quantizing the networks to low bit precision in this work. It allows reduced memory footprint, faster inference time and lower power consumption when



(b) Top-1 accuracy of low-bit MobileNetV2

Figure 1: (a) Illustration of target distributions by different regularization methods. KURE [31] encourages the overall distribution to be uniform, while our bin regularization (BR) encourages each quantization bin distribution to be sharp. (b) Top-1 accuracy of low-bit MobileNetV2 with different regularization methods.

deploying CNN models on the edge devices.

Usually, the network accuracy decreases and the hardware performance increases as bit precision decreases. Recent network quantization methods have shown promising performance for 8-bit quantization by post-training quanti-

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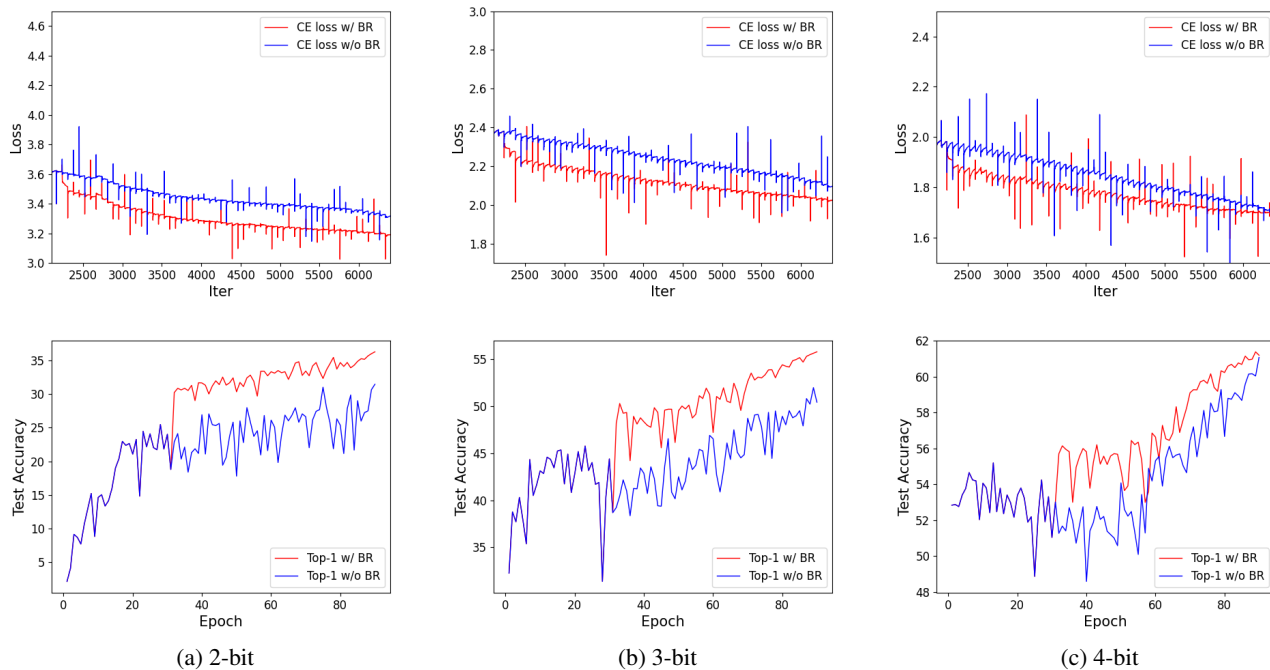


Figure 5: Loss curves and validation accuracies with QAT for different bit widths on MobileNetV3-Small.

Method	W/A	Acc@1 (%)	Acc@5 (%)
Full-precision	32/32	71.8	90.2
LSQ [12]	4/4	69.5	89.2
LSQ [12] + BR (Ours)	4/4	70.4	89.4
TQT [19]	4/4	63.8	85.2
TQT [19]+L2	4/4	65.8	86.7
TQT [19]+BR(Ours)	4/4	67.5	87.7

Table 9: Performance comparisons with other QAT methods on MobileNetV2.

4.7. Training Convergence

Figure 5 presents the loss and accuracy curves during training the low-precision MobileNetV2. With our bin regularization method, the task loss (i.e., CE loss) can be further decreased and the final accuracy can be improved.

4.8. Hardware-Friendly Implementation

Most of the existing quantization methods adopt some relaxations in order to maintain the high accuracy of quantized networks (e.g., the first and last layers are not quantized or quantized to 8 bits). We follow [19, 21] to re-implement a hardware-friendly quantization baseline by power-of-2 step size, quantizing the first and last layer with the same low bit-widths, quantizing both inputs and outputs

of element-wise addition modules, quantizing bias to 8 bits, merging BN to the previous linear layer during training. Table 9 shows that our bin regularization still can improve significantly over the hardware-friendly quantization baseline (e.g., +3.7% over TQT for 4-bit MobileNetV2). We believe that our work can promote the applications of network quantization for different hardware platforms and accelerators in practical industrial scenarios.

5. Conclusion

In this work, we address the neural network quantization problem, especially for low-precision networks. Instead of constraining the overall data distribution, we propose to encourage each quantization bin to be as sharp as possible. By constraining the data falling into each bin to be close to the target quantized value, we can reduce the quantization error at a fine-grained level. We evaluate our method on multiple popular network architectures and achieve the state-of-the-art accuracy on ImageNet classification for 4-bit, 3-bit, 2-bit quantization. We believe that the proposed idea can inspire new insights for the network quantization problem and promote efficient model deployment on resource-limited devices in practical applications.

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